Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/07/2015
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	05/07/2008	**End Date:**	02/28/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 2/28/14 per NSSC information (Ed., 5/1/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang, Zexin (University of Pennsylvania)		
Grant/Contract No.:	NNX08AO80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This proposal is for continuation of Low Volume Fraction Entropically Driven Colloidal Assembly (see "Colloidal Assembly in Entropically Driven, Low-Volume-Fraction Binary Particle Suspensions," Task Book Archives, FY2001 record), an experimental program that was previously peer-reviewed and accepted for flight on the International Space Station (ISS). The instrumentation at the heart of the original flight proposal for ISS was the Light Microscopy Module (LMM). Related complex fluid measurements have also been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight experiments, and newer measurements are planned for the next BCAT mission. Since the original proposal was submitted and approved, the NASA budget has been in flux and soft matter science has evolved too. In response to these developments the present investigations are leaner, with smaller budgets, and the science aims have also been refined to take advantage of important breakthroughs in the field. Herein we describe these complex fluid experiments. In these investigations we will be joining forces with several colleagues from Europe to perform collaborative experiments within the ESA and NASA facilities.
The goal of our research over the next few years will be to identify the most appropriate of these projects for flight experiment. The overall rationale for these investigations is that in microgravity, the elimination of particle sedimentation effects creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape and volume fraction, and interparticle interactions are the primary determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and also to learn about the effects of gravity in soft matter science that are masked in ground-based experiments.

The bulk of our research centers on the physics of complex fluids, an emerging subfield of condensed matter physics. Complex fluids are soft materials such as colloidal suspensions, emulsions, polymer solutions, membranes, and mixtures thereof, whose structure and dynamics are strongly influenced by entropy and by relatively weak mechanical forces.

In this proposal we focus on colloids and mixtures of colloidal particles with other soft materials. These materials are intriguing from both fundamental and practical points of view. Soft suspensions exhibit a wide range of structures, both ordered and disordered, with correlations that often extend to the size of the system. Furthermore they exhibit a wide range of dynamics, which can often be tied closely to the structure of the system. Their rich phenomenology derives from the interplay of physical, chemical, and hydrodynamic mechanisms whose realization provides a unique opportunity for the study of statistical mechanics in classical many-body systems. The experiments proposed emphasize fundamental science in complex fluids, particularly colloidal suspensions.

The Phase 2 version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles and additional smaller constituents (either a small colloidal particle, a polymer or a micelle). The present studies will continue with some of the original experiments and refinements of these measurements. The focus of these measurements reflects the evolution of the complex fluid field as well as some critical new developments in our laboratory. Our proposed research will elucidate a wide-range of phenomena in condensed matter physics including the phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite size colloidal droplets, jamming in 3D, crystallization and melting mechanisms of colloidal liquid-crystals, and aging of filamentous networks.

International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands; Prof. Marzio Giglio, Dipartimento di Fisica, Università degli Studi di Milano, Milan, Italy; Prof. Gerard Wegdam, Van der Waals-Zeeman Institute, University of Amsterdam, Amsterdam, The Netherlands.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research in fluid physics and transport phenomena can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.

The Advanced Colloids Experiment-Heated-1 (ACE-H-1) experiment consists of a series of samples containing dense suspensions of temperature-sensitive microscopic spheres, which are used to study the transition from an ordered crystalline packing to a disordered glassy packing. Regulating the temperature enables control of particle size, sample packing fraction, and allows for variance of particle size uniformity between samples, that will enable the control of disorder in the samples. Further, the goal is understanding how disorder affects the melting and freezing of materials.
• The microscopic process of the transition from a crystal to a glass has been referred to as one of the deep mysteries of our time. A better understanding of this phenomenon will lead to better control in synthesizing glass forming materials, resulting in improved mechanical properties in these materials.

• The crystal-to-glass transition is induced over a series of samples by including and increasing the fraction of small, randomly distributed dopant particle inclusions in a dense packing on uniformly sized colloidal microspheres. The colloidal systems, in particular, are well suited for the investigation of the microscopic structural and dynamic mechanisms that underlie this crossover. These particles have diameters which are sensitive to temperature. This property allows the use of heating and cooling to change the sample phase from crystal/glass, to fluid, and back. The particles are also fluorescent so that their position and motion can be tracked from microscope videos taken throughout the experiment.

• These experiments provide insight into how increasing disorder in an initially crystalline material can affect melting, freezing, aging, and other structural and dynamic properties.

Previous studies with colloidal systems have found that a small amount of gravity dramatically affects the structure and dynamics of a newly formed crystal (e.g., by eliminating sedimentation issues). These effects were masked by the Earth’s gravitational field in terrestrial experiments. By performing experiments on dense suspensions with varying degrees of imperfection in microgravity, we can similarly better understand the fundamental physics of nucleation, aging, and melting in increasingly disordered materials. Additionally, these experiments will develop the technology for observation and synthesis of new colloidal materials (e.g., meta- and photonic-materials,) and colloid engineering in microgravity.

Task Progress:	Understanding the transition from ordered (crystal) to disordered (glassy) materials will teach ways to alter the macroscopic behavior of disordered materials by frustrating their structural order. Additionally, these experiments will provide fundamental insight into the nature of nucleation, aging, and melting in materials with varying degrees of order. The ACE-H-1 experiment utilizes the Fluids Integrated Rack (FIR) and the Light Microscopy Module (LMM). The FIR is built to accommodate a wide range of experiments. The fluid physics research focuses on complex fluids, interfacial phenomena, dynamics and instabilities, multiphase flows, and phase changes. Investigations range from fundamental research to technology development in support of the NASA exploration missions and include life support, power, propulsion, and thermal control systems. The FIR minimizes up mass by using different modules that can support various types of experiments. The Light Microscopy Module (LMM) is an automated microscope that is being built up in stages. It will allow flexible imaging (e.g., bright field, dark field, phase contrast, confocal microscopy) for physical and biological experiments and can be controlled remotely (commanded from the ground). The LMM accommodates sample change out and fluid containment and includes a glovebox for on-orbit sample manipulation. This unique capability can support a large set of experiments that require visual imaging of a small test sample. ACE-H-1 utilizes two sample cell modules, each with 15 sample wells (30 wells total). • Each well holds around 2 microliters of sample fluid. • Two sample cell modules provide 3 stips of 5 sets of samples (15 samples per sample cell module). 15 good wells of the 30 wells provided by the two sample cell modules are required for mission success More than 15 are desired, time allowing. • All wells contain stainless steel stir bars (magnetic). After an initial mixing, simply heating the samples will probably be all that is needed to remix them. • There are two fill ports for each well. The ports are filled with Kalrez® plugs. Summers Optical Cement F-65 is applied over the plugs. ACE-H-1 Sample Cell Modules are mounted on a temperature controlled Preliminary Advanced Colloids Experiment (PACE) LED Base for installation in the LMM. The colloidal samples will be water-dispersed; thus, they must be kept above 3°C at all times to prevent freezing of the sample and resulting destabilization of colloidal particles and/or fracturing of sample cells. Additionally, these experiments will have a high fraction of colloidal particles, and will thus be very sensitive to sample leakage and/or evaporation. This should be prevented and/or kept to an absolute minimum. A previous mission on the ISS, BCAT-5 (sample 10), executed a related experiment with thermo-responsive poly(N -isopropylacrylamide) (pNIPA) particles. Here, the macroscopic structure of a 3D crystalline packing of monodisperse pNIPA particles was probed using light scattering. Ideally, temperature fluctuations in the cabin would have varied the volume fraction of the packing periodically, allowing observations of melting/crystallization in the packing. Publications acknowledging the NASA grant are listed in the Bibliography section. NOTE that project continues as "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)"; grant NNX13AL27G.

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